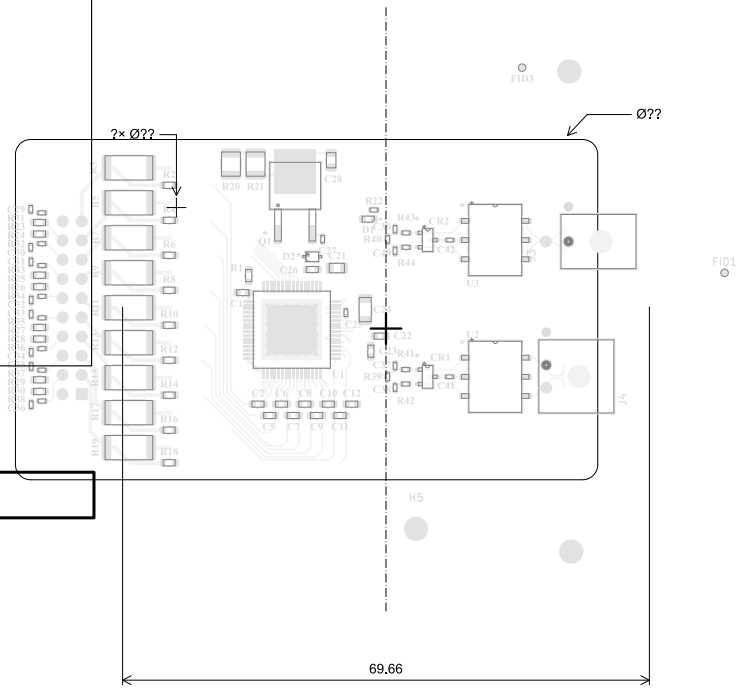
BMS_SLAVE Fabrication Document

Layer Stack Legend

Impedance Table

Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

4) SURFACE FINISH: IMMERSION GOLD

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLUE.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY BLUE DEVIL MOTOR SPORTS.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	77.000 × 45.000 mm
BOARD THICKNESS	1.581 mm
TRACE WIDTH	0.200 mm
TRACE TO TRACE	-0.000 mm
MIN. HOLE (PTH)	0.200 mm
MIN. HOLE (NPTH)	3.050 mm
ANNULAR RING	0.150 mm
COPPER TO HOLE	0.254 mm
COPPER TO EDGE	0.250 mm
HOLE TO HOLE	0.254 mm

11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.

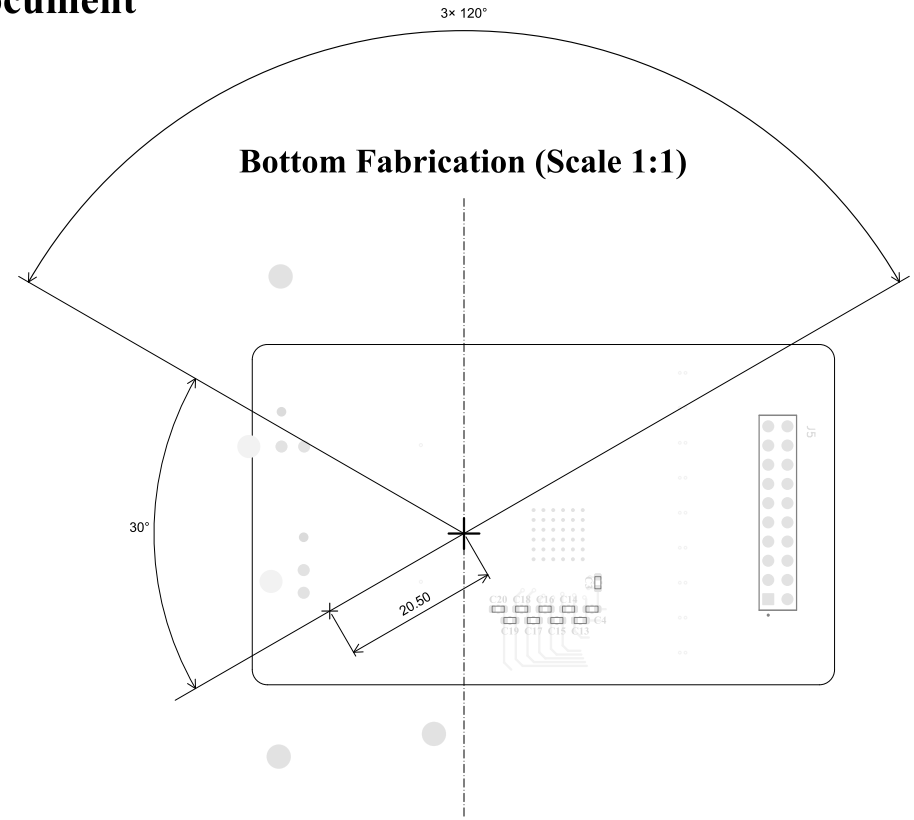
12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: 30e9a8d
	Board Name: BMS_SLAVE		Project Name: BMS Slave	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: bms_slave.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 1 of 10

BMS_SLAVE Fabrication Document

Bottom Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

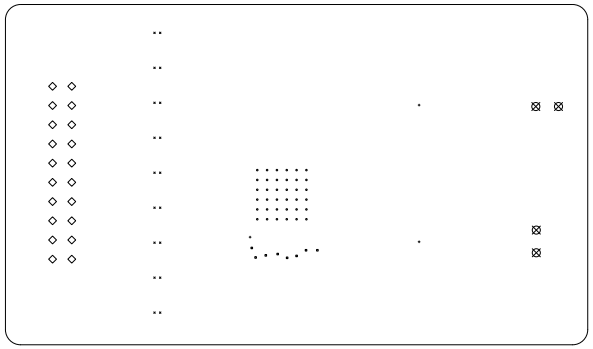
blue devil MOTORSPORTS	Comments:	Company: Blue Devil Motor Sports	Variant: PRELIMINARY	Git Hash: 30e9a8d
		Board Name: BMS_SLAVE	Project Name: BMS Slave	
	Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: bms_slave.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13 / Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4 / Sheet: 2 of 10

BMS_SLAVE Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	18	0.20mm (7.87mils)	PTH	Round	L1 – L4	Via
○	36	0.20mm (8.00mils)	PTH	Round	L1 – L4	Pad
+	3	0.25mm (9.84mils)	PTH	Round	L1 – L4	Via
□	8	0.30mm (11.81mils)	PTH	Round	L1 – L4	Via
◇	20	1.04mm (40.94mils)	PTH	Round	L1 – L4	Pad
⊠	4	1.07mm (42.00mils)	PTH	Round	L1 – L4	Pad
	Total 89					

Drill Drawing L1 - L4 (Scale 1:1)



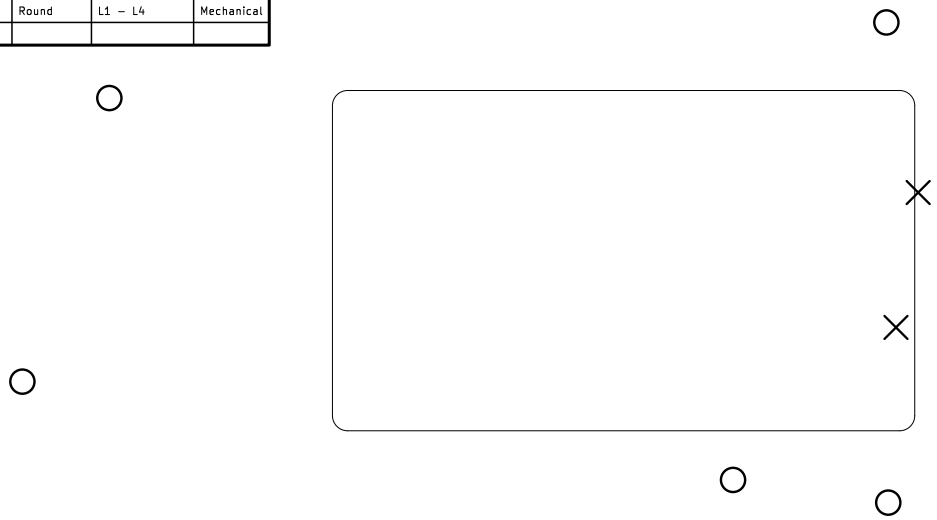
Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: 30e9a8d
	Board Name: BMS_SLAVE		Project Name: BMS Slave	
Sheet Title: Drill Drawing (L1 - L4)	File Name: bms_slave.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 3 of 10

BMS_SLAVE Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	2	3.05mm (120.00mils)	NPTH	Round	L1 – L4	Mechanical
○	5	3.20mm (125.98mils)	NPTH	Round	L1 – L4	Mechanical
	Total 7					

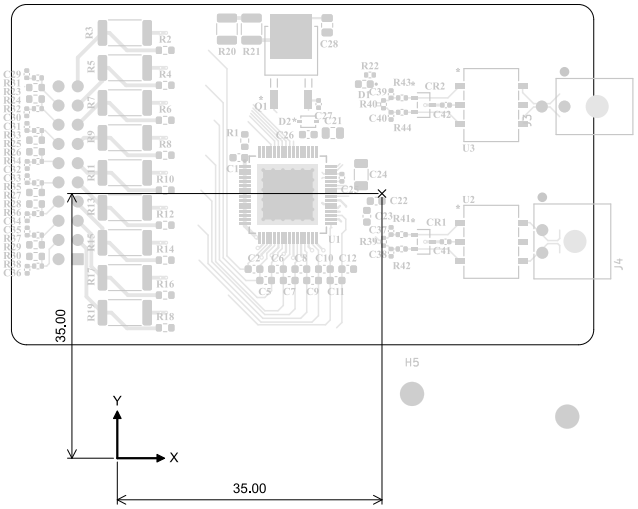
Drill Drawing L1 - L4 (Scale 1:1)



Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: 30e9a8d	
	Board Name: BMS_SLAVE		Project Name: BMS Slave		
Sheet Title: Drill Drawing (L1 - L4)	File Name: bms_slave.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)	
Sheet Path:		Reviewer:	Size: A4	Sheet: 4 of 10	

BMS_SLAVE Fabrication Document

Top Test Points (Scale 1:1)



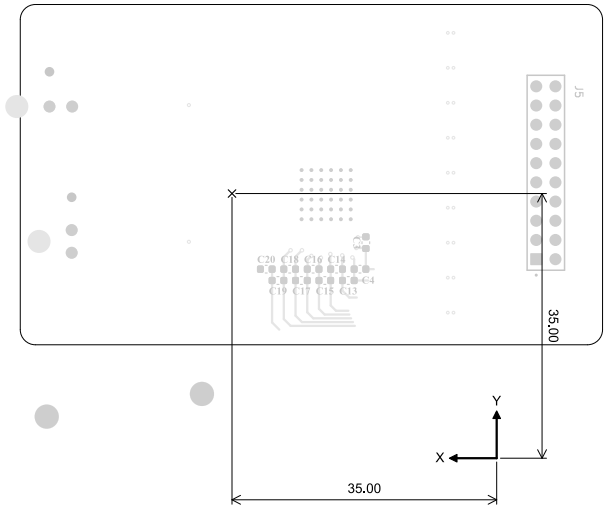
Ref.	Net	X [mm]	Y [mm]
TP1	CVDD	-19.25	56.40
TP2	DVDD	-19.25	52.35
TP3	AVDD	-15.20	56.40
TP4	LDOIN	-15.20	52.35
TP5	VCOLL	-11.15	56.40
TP6	LDOIN	-11.15	52.35
TP7	TSREF	-7.10	56.40
TP8	GND	-7.10	52.35
TP9	T_P1	-19.25	66.50
TP10	GND	-19.25	62.45
TP11	U1_COMLP	-25.30	66.50
TP12	U1_COMLN	-25.30	62.45
TP13	U1_COMHP	-25.30	58.40
TP14	U1_COMHN	-25.30	54.35

All dimensions are in millimeters unless otherwise specified.

Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: 30e9a8d
	Board Name: BMS_SLAVE		Project Name: BMS Slave	
Sheet Title: Top Test Points (Scale 1:1)	File Name: bms_slave.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 5 of 10

BMS_SLAVE Fabrication Document

Bottom Test Points (Scale 1:1)

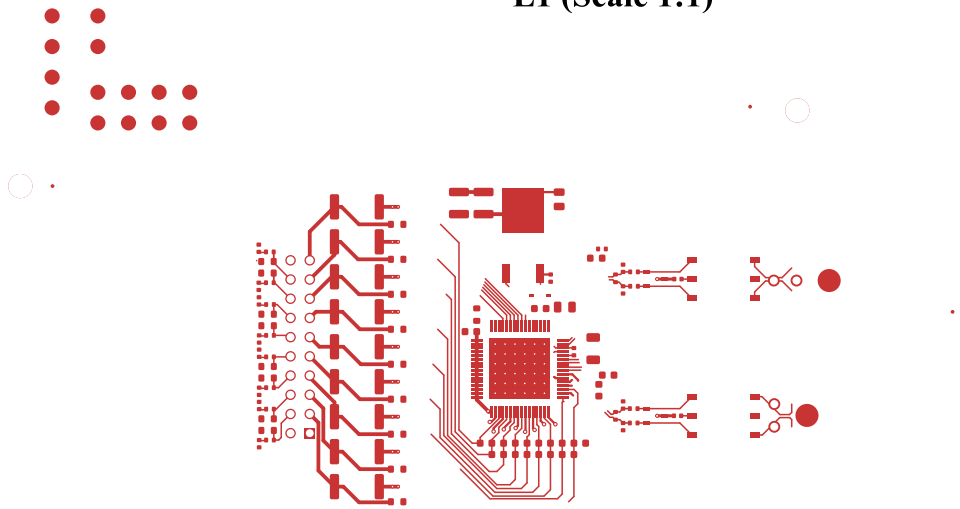


All dimensions are in millimeters unless otherwise specified.

Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: 30e9a8d
	Board Name: BMS_SLAVE		Project Name: BMS Slave	
Sheet Title: Bottom Test Points (Scale 1:1)	File Name: bms_slave.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 6 of 10

BMS_SLAVE Fabrication Document

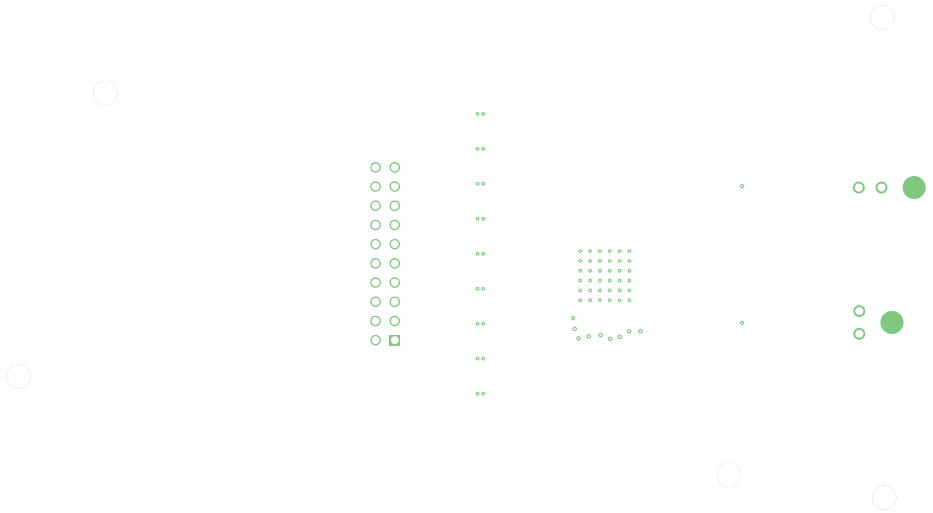
L1 (Scale 1:1)



Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: 30e9a8d
	Board Name: BMS_SLAVE		Project Name: BMS Slave	
Sheet Title: L1 (Scale 1:1)	File Name: bms_slave.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 7 of 10

BMS_SLAVE Fabrication Document

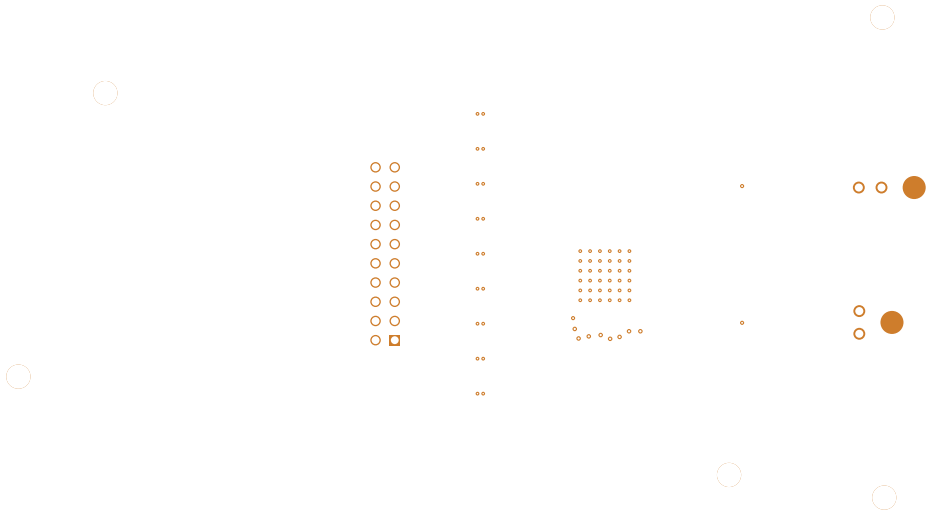
L2 (Scale 1:1)



Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: 30e9a8d
	Board Name: BMS_SLAVE		Project Name: BMS Slave	
Sheet Title: L2 (Scale 1:1)	File Name: bms_slave.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 8 of 10

BMS_SLAVE Fabrication Document

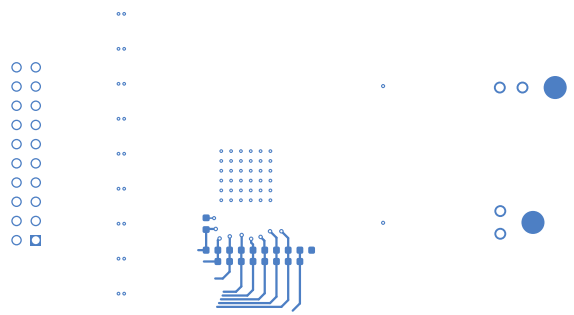
L3 (Scale 1:1)



Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: 30e9a8d
	Board Name: BMS_SLAVE		Project Name: BMS Slave	
Sheet Title: L3 (Scale 1:1)	File Name: bms_slave.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 9 of 10

BMS_SLAVE Fabrication Document

L4 (Scale 1:1)



Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: 30e9a8d
	Board Name: BMS_SLAVE		Project Name: BMS Slave	
Sheet Title: L4 (Scale 1:1)	File Name: bms_slave.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 10 of 10